K_c & K_p expression

Write the K_c or K_p expression and the units for the following chemical equations

$2CH_3OH(aq) + (COOH)_2(aq) \rightleftharpoons (COOCH_3)_2(aq) + 2H_2O(l)$

K_c = units =

$H_2O(g) + C(s) \rightleftharpoons H_2(g) + CO(g)$

K_p = units =

Note:

- K_c and K_p are calculated in terms of concentration and partial pressure respectively, <u>not number of mol</u>
- K_c and K_p are <u>only</u> affected by temperature

ICE Table

Complete the following ICE tables

example 1	$CH_4(g)$	+	$H_2O(g)$	$\rightleftharpoons$	$CO(g)$	+	$3H_2(g)$
initial / atm	3.0		1.0		0		0
change / atm							
eqm / atm					0.5		

example 2	$CH_4(g)$	+	$H_2O(g)$	$\rightleftharpoons$	$CO(g)$	+	$3H_2(g)$
initial / mol	25		15		0		0
change / mol							
eqm / mol					x		

Note:

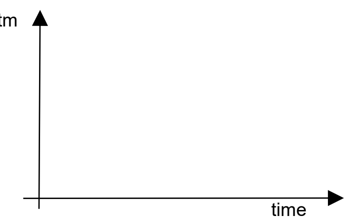
- ICE table by default is calculated in terms of mol.
  It can be calculated in terms of conc. or pressure only if the quantities are proportional to number of mols

Reversible reactions graphs

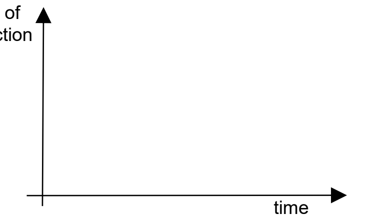
3 atm of $CH_4(g)$ and 1 atm of $H_2O(g)$ were mixed in a sealed vessel and reacts according to the following equation:

$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$

(i) Sketch a graph to show how partial pressures of $CH_4(g)$ and and $H_2(g)$ changes



(ii) Sketch a graph to show the rates of the forward and backwards reaction



Explaining shift in position of equilibrium (POE)

$2NO_2(g) \rightleftharpoons N_2O_4(g)$ $\Delta H < 0$.

A sealed gas syringe contains a mixture of $NO_2(g)$ and $N_2O_4(g)$ at equilibrium.
Explain how the following changes results in a shift in position of equilibrium.

change	explanation by Le Chatelier's Principle	thought process by considering Q vs K
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ more $NO_2(g)$ is added, while keeping volume constant	When more NO_2 is added, the $[NO_2]$ increases. By LCP, the system counteracts the increase in $[NO_2]$ by favoring the ______________ reaction that consumes NO_2. Position of equilibrium shifts ____________	$Q_c = \frac{[N_2O_4]}{[NO_2]^2}$
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ temperature increases	When temperature increases, by LCP, the system counteracts the increase in temperature by favoring the ______________ ______________ reaction that ____________________________________ Position of equilibrium shifts ___________	-
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ syringe is compressed, while keeping temperature constant	When volume of the syringe decreases, total pressure increases. By LCP, the system counteracts the increase in pressure by favoring the _____________________ reaction that ____________________________________ Position of equilibrium shifts ___________	$Q_p = \frac{(P_{N_2O_4})}{(P_{NO_2})^2}$
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ Inert Ar(g) is added, while keeping volume constant		$Q_p = \frac{(P_{N_2O_4})}{(P_{NO_2})^2}$

K_c & K_p expression

Write the K_c or K_p expression and the units for the following chemical equations

$2CH_3OH(aq) + (COOH)_2(aq) \rightleftharpoons (COOCH_3)_2(aq) + 2H_2O(l)$

$$K_c = \frac{[(COOCH_3)_2]}{[CH_3OH]^2\,[(COOH)_2]}$$

units = $mol^{-2}\ dm^6$

$H_2O(g) + C(s) \rightleftharpoons H_2(g) + CO(g)$

$$K_p = \frac{(p_{H2})(p_{CO})}{(p_{H2O})}$$

units = atm or Pa

Note:

- K_c and K_p are calculated in terms of concentration and partial pressure respectively, not number of mol
- K_c and K_p are only affected by temperature

ICE Table

Complete the following ICE tables

example 1	$CH_4(g)$	+ $H_2O(g)$	$\rightleftharpoons$ $CO(g)$	+ $3H_2(g)$
initial / atm	3.0	1.0	0	0
change / atm	−0.5	−0.5	+0.5	−1.5
eqm / atm	2.5	0.5	0.5	1.5

example 2	$CH_4(g)$	+ $H_2O(g)$	$\rightleftharpoons$ $CO(g)$	+ $3H_2(g)$
initial / mol	25	15	0	0
change / mol	−x	−x	+x	+3x
eqm / mol	25 − x	15 − x	x	3x

Note:

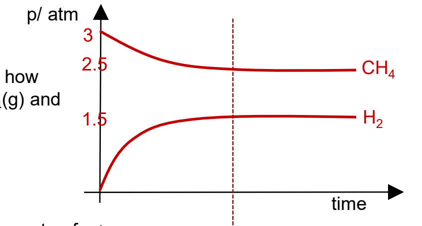
➢ ICE table by default is calculated in terms of mol. It can be calculated in terms of conc. or pressure only if the quantities are proportional to number of mols

Reversible reactions graphs

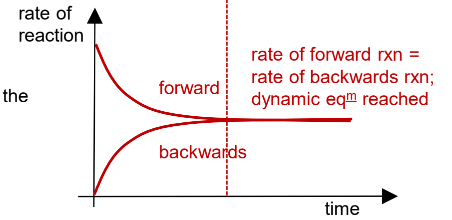
3 atm of $CH_4(g)$ and 1 atm of $H_2O(g)$ were mixed in a sealed vessel and reacts according to the following equation:

$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g)$

(i) Sketch a graph to show how partial pressures of $CH_4(g)$ and $H_2(g)$ changes



(ii) Sketch a graph to show the rates of the forward and backwards reaction



Explaining shift in position of equilibrium (POE)

$2NO_2(g) \rightleftharpoons N_2O_4(g)$ $\Delta H < 0$. (exo)

A sealed gas syringe contains a mixture of $NO_2(g)$ and $N_2O_4(g)$ at equilibrium.
Explain how the following changes results in a shift in position of equilibrium.

change	explanation by Le Chatelier's Principle	thought process by considering Q vs K
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ more $NO_2(g)$ is added, while keeping volume constant	When more NO_2 is added, the $[NO_2]$ increases. By LCP, the system counteracts the increase in $[NO_2]$ by favoring the forward reaction that consumes NO_2. Position of equilibrium shifts right	$Q_c = \frac{[N_2O_4]}{[NO_2]^2 \uparrow\uparrow} < K_c$ hence the position of equilibrium will shift right
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ + heat temperature increases	When temperature increases, by LCP, the system counteracts the increase in temperature by favoring the backwards endothermic reaction that consumes heat Position of equilibrium shifts left	-
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ syringe is compressed, while keeping temperature constant	When volume of the syringe decreases, total pressure increases. By LCP, the system counteracts the increase in pressure by favoring the forward reaction that produces less number of mol of gas Position of equilibrium shifts right	$Q_p = \frac{(P_{N_2O_4})\uparrow}{(P_{NO_2})^2 \uparrow\uparrow} < K_c$ hence the position of equilibrium will shift right
$2NO_2(g) \rightleftharpoons N_2O_4(g)$ Inert Ar(g) is added, while keeping volume constant	Partial pressure of both NO_2 and N_2O_4 remains constant. Hence there is no shift in position of equilibrium.	$Q_p = \frac{(P_{N_2O_4})}{(P_{NO_2})^2} = K_c$ hence the position of equilibrium will not shift